

# 1 PCC Configuration DNC Stand-alone

## 1.1 Usage

You use DNC as a stand-alone solution in PCC in order to perform uploads and downloads of DNC programs directly on the machine without any AIP terminal functions.

The PCC configuration (pcc.exe) as well as the file driver configuration (dnc\_dr\_01.dll), which exchanges files in PCC-DIF format with the machine, are described below.

The following versions are the minimum requirement:

| Program       | Minimum version |
|---------------|-----------------|
| pcc.exe       | 7.2.2.79        |
| pccdll.dll    | 7.2.2.12        |
| dncb.dll      | 7.2.1.1         |
| dnc_dr_01.dll | 7.2.1.1         |

## 1.2 Configuration

### 1.2.1 PCC Configuration

|           |         |
|-----------|---------|
| File      | pcc.exe |
| INIT file | pcc.ini |

The DLL "pccDll.dll" has to be configured as BusDLL in PCC. In addition, PCC is configured as stand-alone solution with specification of the Hydra server and port as well as the PCC terminal number. Communication with a terminal (gateway communication) must be deactivated. In the "BLADES" section, the DNC blade "DNCB.DLL" must be configured. In the "FILTER" section, the commands RES\_UPLOAD and RES\_DOWNL are indicated so that they may be processed in the DNC blade.

```
; pccd11.dll has to be configured as BusDll here  
[DLL]
```

```
BusDLL=pccD11.dll
```

```
[SYSTEM]
SYSTRAY_TIME=2
ALIVE_SIGNAL=60
viewmode=2
RUNSTANDALONE=ON
AutoTerminate=20
;;parameters= --AlwaysReloadScript
TRANSMISSION_RETRIES=2
TRANSMISSION_FAILED=CONTINUE
QUEUESIZE=1000000
;TRANSMISSION_FAILED=STOP
NORMALQUEUECYCLE=1000
FASTQUEUECYCLE=1000

; Entry of server and user no. of PCC
[WSK]
; HOSTNAME OR IP-ADDRESS OF SERVER
Host=HYDRASSERVER
; PORT OF SERVER FOR CLIENT COMMUNICATION
; 1. SYSTEM 10000
Port=10000
; UNIQUE USR ID (TERMINAL ID)
User=9

[GateWay-Communication]
; GATEWAY SHOULD BE DEACTIVED WHEN PCC IS RUNNING STANDALONE
Active=false
;Port=9003

[Tracing]
; 5 FOR TESTING ENVIRONMENT
; 0 FOR PRODUCTIVE ENVIRONMENT
TraceLevel=5

[Server-Communication]
; COMMUNICATION WITH SERVER VIA EVCOM
; NECESSARY FOR SENDING INFORMATION FROM SERVER TO CLIENT
; MUST BE ACTICE (= 1) WHEN BEVERAGE SOLUTION
; PORTS MUST BE UNIQE WHEN MULTIPLE INSTANCES ON SAME PC ARE ACTIVE
Active=1
Port=9005

[blade_value_dncb]
;Refreshrate=10000
[blade_value_pdv]
TraceLevel=1
Refreshrate=50000

[BLADES]
; THESE BLADES HAVE TO BE LOADED
BLADE_1=.\blades\DNCB.DLL
;BLADE_2=.\blades\PDV.DLL
; Section for filtering which dialog data are sent to DNC blade
```

```
[FILTER]
; FILTER FOR BLADE [DNCB]
DNCB= RES_UPLOAD, RES_DOWNL
```

## 1.2.2 PCCDLL Configuration

|          |            |
|----------|------------|
| File     | pccdll.dll |
| INI file | pccdll.ini |

The file driver dnc\_dr\_01.dll is activated in pccdll.ini as follows:

```
[SERVICE]
tracing=5
;DNC_BLOCKED_PDV=ON
DNC_TIMEOUT=240
DLL-VERSION=
Bypass_PDV=ON
```

```
[DRIVER_1]
driver=dnc_dr_01.dll
```

```
;[DRIVER_2]
;Next driver possible here)
```

```
;[DRIVER_3]
;Next driver possible here)
```

## 1.2.3 DNC Blade Configuration (in PCC)

|          |                                            |
|----------|--------------------------------------------|
| File     | dncb.dll                                   |
| INI file | No INI file is required for the DNC blade. |

### General information on the DNC blade

The DNC blade is located in the "Blades" directory which is found in the PCC application directory:

**Example:** D:\pcc\blades\  
**pcc spool directory** D:\pcc\spool\

The DNC blade creates own directories in the application directory of PCC:

- The working directory for upload and download files  
**\pcc\_dnc\_dir\**  
is created in the installation directory.
- If offline or in case of an error, it is not possible to transport files to the server. For this reason, these upload files are stored in the directory  
**\pcc\_dnc\_dir\_offline\**  
under the installation directory.

The file name structure of a file in this directory is

**xxxxxxx-YYMMDD-hhmmss\_Filename.Extension**

and is composed of the following elements:

- Machine number, 8 digits (if the machine has less digits, hyphens are prefixed (----xxxx))
- Date and time
- Original file name with file extension

The valid optional extension is used as the file extension.

## 1.2.4 File Driver Configuration

|                  |               |
|------------------|---------------|
| <b>File</b>      | dnc_dr_01.dll |
| <b>INIT file</b> | dnc_dr_01.ini |

The file driver dnc\_dr\_01.dll is configured in dnc\_dr\_01.ini as follows. The meanings of the tokens (**INTERPRETER\_...**) are defined in Section 1.4 "Use without Interpreter".

**The following sections are contained in the INI file (dnc\_dr\_01.ini).**

**Example of an instance for machine 6125**

```
[SERVICE]
info=
intervall=500
testmode=0
tracing=1
TraceLevel=5
ExecuteQueue=0
DNC_SETERROR=OFF
DNC_SETTIMEOUT=OFF
DNC_TRANSFERERROR=OFF
DNCPROTOKOLL=ON
; 1. Instanc Section
; For other machines, additional instances have to be created.
; Each instance is provided with an own designation. [xxx], [yyy]
[DNC1]
;TIMEOUT-DELETE-DOWNLFILES=1
;CLR_AFTER_DOWNLOAD=OFF
```

```
; Use with interpreter
; Configuration for interpretation of function
; Path and file name of interpreter program
;INTERPRETER=D:\4711_interpreter\interpreter.exe
```

```
; Use without interpreter
INTERPRETER_ERRCODE_TOKEN=ERR
INTERPRETER_ERRTEXT_TOKEN=TEXT
INTERPRETER_DOWNLOAD_TOKEN=DNC-LOAD
INTERPRETER_PROG_TOKEN=PROGFILE
INTERPRETER_COMMENT_CHAR=;
; Waiting time in seconds until download file is written into upload/download
directory.
; This can be omitted in file transfer.
```

```

; If parameter is not set, the file is written immediately.
;DOWNLOAD_WAIT_PERIOD=10
; The upload file is stored on the server under the resource name.
;;INTERPRETER_FILENAME_AS_PROG=OFF
; This is only required if the DNC blade directory was changed,
; so that the driver knows the DNC blade directory.
;;DNC_BLADE_DIR=d:\pcc\pcc_dnc_dir\
;Upload Path
;If configured, this path is used as upload directory
;UPL-PATH_6125=d:\dnc_upl\
; Machine download and upload path; here, as an example for
; machine 6125, it is d:\dnc_dir\
D:DNC_6125=d:\dnc_dir\
; So that the driver scans the upload/download directory
; Parameter POLL must be set to 1.
POLL=1
; Parameter POLL_I=1000 means that every 1000 milliseconds
; the directory is polled.
POLL_I=1000

```

**Please note:**

- All lines marked with a semicolon are defined as comment lines.
- The DNC driver is located in the directory of the PCC and DNC blade.

## 1.3 Use with Interpreter

The interpreter (configured with the value **INTERPRETER** in the INI file of the file driver) is activated with the following parameters

|            |                                                                                |
|------------|--------------------------------------------------------------------------------|
| Parameter1 | File name of file sent by the machine                                          |
| Parameter2 | File name of file into which the interpreter writes the command to be executed |

Both file names are transferred to the interpreter with the full path.

### Configuration example

```

INTERPRETER=D:\4711_interpreter\interpreter.exe
D:DNC_6125=d:\dnc_dir\

```

The machine file is provided as test.prg in directory d:\dnc\_dir\.

The interpreter is now activated as follows:

```
D:\4711_interpreter\interpreter.exe d:\dnc_dir\test.prg d:\dnc_dir\test.cmd
```

The interpreter writes the relevant DNC command in the command file (as PDM command).

- In the case of a download command, the command DLG=RES\_DOWNL is written in the command file with specification of the resource (RES=...) to be loaded:

```
DLG=RES_DOWNL|RES=4521580105000|
```

- In the case of an upload command, the command DLG=RES\_UPLOAD is written in the command file with specification of the resource (RES=...) to be modified or created:

```
DLG=RES_UPLOAD | RES=4521580105000 |
```

## 1.4 Use without Interpreter

### 1.4.1 Structure of DNC Upload File

To enable the internal interpreter to recognize an upload request, the following tokens have to be configured:

| Configuration                   | Description                                                                                 |
|---------------------------------|---------------------------------------------------------------------------------------------|
| INTERPRETER_COMMENT_CHAR=;      | Semicolon is defined as comment character here                                              |
| INTERPRETER_PROG_TOKEN=PROGFILE | Defines the resource designation under which the DNC upload file is assigned on the server. |

An upload file has to include the specification of the resource in the header.

Example:

```
; PROGFILE=20130204
```

```
...Programmcode...
```

```
...Programmcode...
```

```
...Programmcode...
```

If, however, no resource is indicated in the file, the DNC blade will automatically create a resource with a designation (max. 20 digits) derived from the **machine number** (8 digits) and the time stamp. The format of the resource designation is as follows:

```
xxxxxxxxYYMMDDhhmmss.
```

If the machine number has less than 8 digits, the machine number is prefixed with hyphens.

Example:

```
----4711130204161000
```

Upload of machine 4711 on 2/4/2013 at 4:10:00 PM.

### 1.4.2 Structure of DNC Download File Request

To enable the internal interpreter to recognize a download request, the following tokens have to be configured:

| Configuration                   | Description                                      |
|---------------------------------|--------------------------------------------------|
| INTERPRETER_COMMENT_CHAR=;      | Semicolon is defined as comment character here   |
| INTERPRETER_PROG_TOKEN=PROGFILE | Defines the resource designation under which the |

|                                             |                                                        |
|---------------------------------------------|--------------------------------------------------------|
|                                             | DNC upload file is assigned on the server.             |
| INTERPRETER_DOWNLOAD_TOKEN= <b>DNC-LOAD</b> | Defines the identification for a DNC download command. |

A download file has to include the specification of the resource **and the download token** in the header.

Example:

```
;DNC-LOAD
;PROGFILE=9999000
```

### 1.4.3 Structure of Error File (instead of Download File) for the Machine

If a download cannot be executed, an error file is generated for the machine (in the driver). The driver generates the error file with the download resource name and the file extension of a valid DNC file.

In order for an error file to be generated, the following tokens have to be configured:

| Configuration                           | Description                                                                                 |
|-----------------------------------------|---------------------------------------------------------------------------------------------|
| INTERPRETER_COMMENT_CHAR=;              | Semicolon is defined as comment character here                                              |
| INTERPRETER_PROG_TOKEN= <b>PROGFILE</b> | Defines the resource designation under which the DNC upload file is assigned on the server. |
| INTERPRETER_ERRCODE_TOKEN= <b>ERR</b>   |                                                                                             |
| INTERPRETER_ERRTEXT_TOKEN= <b>TEXT</b>  |                                                                                             |

The file is generated based on the configuration. The file has the following structure (configuration above used in this example):

```
;PROGFILE=20130204
;ERR=3218 TEXT=Resource locked
```

### 1.4.4 RS232 Machine Interfacing

Possible as from version dnc\_dr\_01.dll 7.2.1.2

RS232 machine interfacing is activated with the following parameters.

For the machine with RS232 interfacing, the parameter must be entered in the relevant DNC instance.

**COM=3,9600,8,N,1**

**RS232\_CONNECT=ON**

**RS232\_ALIVE\_CYCLETIME=2**